

Data Preparation

Disk N

Disk 1



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Preprocessing

0.6	0.7	-1.2	0.7
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Filter invalid values

10	11	Nan	13
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10	11	12	13
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Linear interpolation

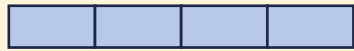
1	2	4	6
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0	0.2	0.6	1
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Min-max normalization

Historical Deviation Analysis

Metric Reconstruction

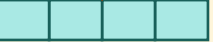


Encoder

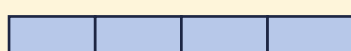


DecoderG

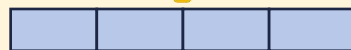
DecoderD



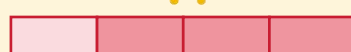
Score Calculation



+

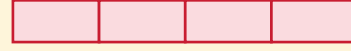


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Anomaly Identification

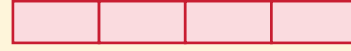
threshold



anomaly score



threshold



anomaly score



Peer Component Comparison

Feature Extraction

0	0.2	0.6	1
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maximum

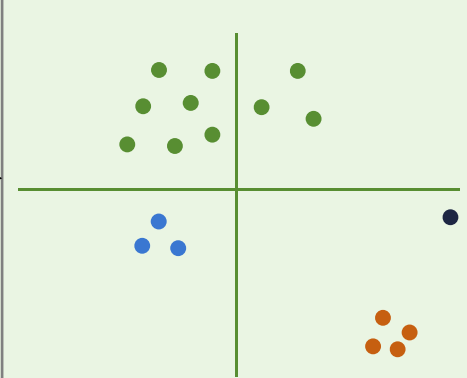
minimum

mean

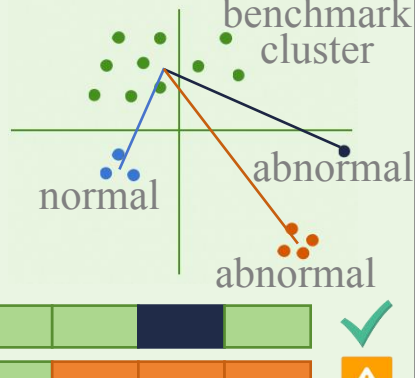
median

...

Component Clustering



Outlier Filtering



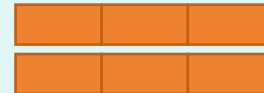
Pattern-Centric Labeling

Knowledge-Driven Filtering

disk_util: increasing
disk_iops: decreasing
...

Anomaly Segment Construction and Representation

Disk K



concat

Disk K feature vector



Sampling and Labeling

Disk K 08:31 - 23:56